

The Security Lottery: Measuring Client-Side Web Security Inconsistencies

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The Security Lottery: Measuring Client-Side Web Security Inconsistencies

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Abstract:

To mitigate a myriad of Web attacks, modern browsers support client-side security policies shipped through HTTP response headers. To enforce these defenses, the server needs to communicate them to the client, a seemingly straightforward process. However, users may access the same site in variegated ways, e.g., using different User-Agents, network access methods, or language settings. All these usage scenarios should enforce the same security policies, otherwise a security lottery would take place: depending on specific client characteristics, different levels of Web application security would be provided to users (inconsistencies). We formalize security guarantees provided through four popular mechanisms and apply this to measure the prevalence of inconsistencies in the security policies of top sites across different client characteristics. Based on our insights, we investigate the security implications of both deterministic and non-deterministic inconsistencies, and show how even prominent services are affected by them.

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